

CS601 Machine Learning Core Formulas & Exam Guide

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Unit 1: Supervised Learning & Linear Regression

Linear Regression hypothesis: $h_{\theta}(x) = \theta^T x$.

Mean Squared Error (MSE) Cost Function: $J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$.

Gradient Descent update rule: $\theta_j := \theta_j - \alpha * \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) * x_j^{(i)}$.

Normal Equation closed-form solution: $\theta = (X^T X)^{-1} X^T y$.

Unit 2: Logistic Regression & Classification

Logistic Regression hypothesis: $h_{\theta}(x) = g(\theta^T x)$ where $g(z) = \frac{1}{1 + e^{-z}}$ is the sigmoid activation function.

Cross-Entropy Loss (Log Loss):

$J(\theta) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)}))]$.

Confusion Matrix Metrics:

- Precision = $TP / (TP + FP)$
- Recall (Sensitivity) = $TP / (TP + FN)$
- F1-Score = $2 * (Precision * Recall) / (Precision + Recall)$
- Specificity = $TN / (TN + FP)$

Unit 3: Support Vector Machines (SVM) & Decision Trees

Support Vector Machines maximize the margin $2 / \|w\|$ between hyperplanes: $\min (1/2) \|w\|^2$ subject to $y_i (w^T x_i + b) \geq 1$.

Kernel Trick maps non-linear data into higher dimensional space:

- Polynomial Kernel: $K(x, z) = (x^T z + c)^d$
- Radial Basis Function (RBF/Gaussian): $K(x, z) = \exp(-\gamma \|x - z\|^2)$.

Decision Trees:

Information Gain = $Entropy(Parent) - \sum (|S_v| / |S|) * Entropy(S_v)$.

Entropy $H(S) = -\sum p_i \log_2(p_i)$.

Gini Impurity = $1 - \sum (p_i)^2$.

Unit 4: Unsupervised Learning & PCA

K-Means Clustering:

1. Initialize k cluster centroids randomly.
2. Assignment Step: Assign each point to the nearest centroid.
3. Update Step: Recompute centroid as the mean of assigned points. Repeat until convergence.

Principal Component Analysis (PCA):

1. Standardize data matrix X .
2. Compute Covariance Matrix $\Sigma = (1/m) X^T X$.
3. Compute eigenvectors and eigenvalues of Σ .
4. Select top k eigenvectors with highest eigenvalues to project into k -dimensional subspace.

